UPIFlow 360 – Digital Payment Intelligence

SQL + Power BI analytics on 5,000 UPI transactions

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5,000

Transactions analyzed



71.48%

Overall success rate



3

Interactive dashboard pages

Project at a glance

INDUSTRY

Digital Payments / FinTech

DATABASE

MySQL | upiflow360

TABLE

UPI_Transactions

DATASET

5,000 records | CSV

TOOLS

MySQL, Power BI, Power Query, DAX

SOURCE

Raw dataset from ChatGPT, cleaned

Problem and Objectives

The problem

Digital payment platforms generate large volumes of transaction data across apps, transaction types, merchant categories, locations, devices and statuses.

This project analyzes that data to understand payment performance, application behavior and transaction reliability.

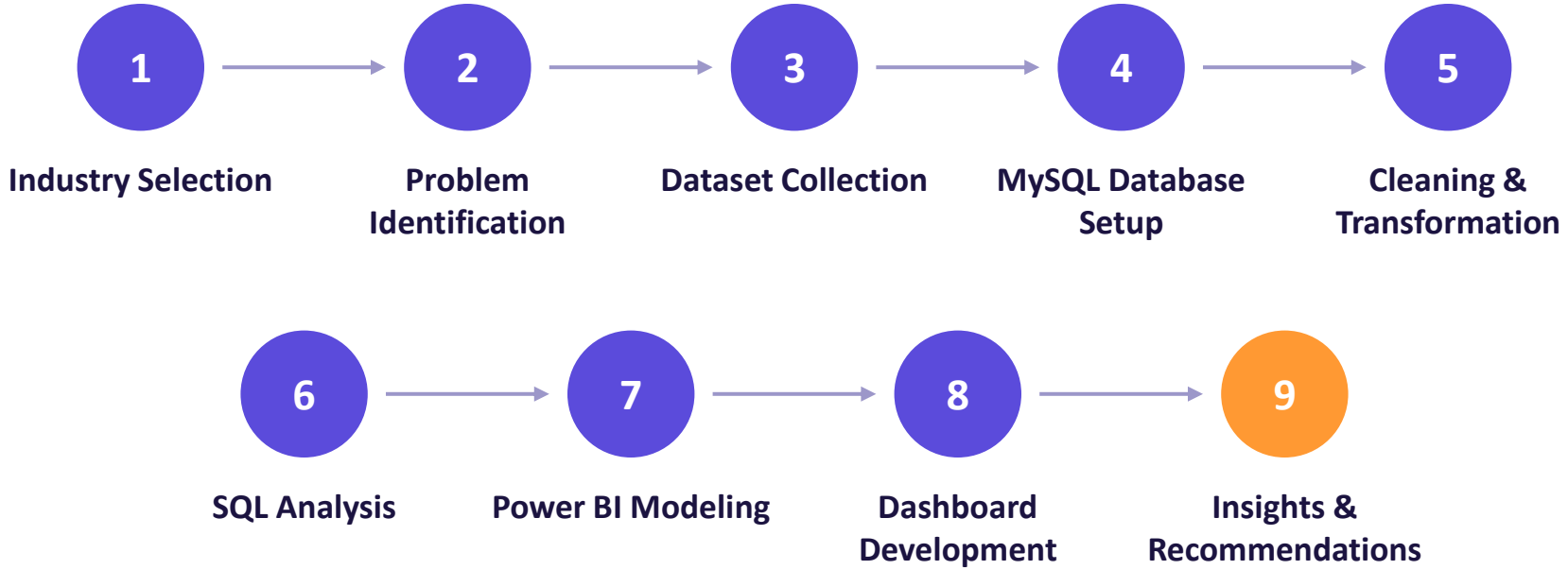
Objectives

1. Monitor transaction volume and value
2. Measure successful, failed and pending transactions
3. Calculate Success, Failure and Pending rates in DAX
4. Compare payment apps by activity and reliability
5. Analyze merchant categories and transaction types
6. Identify major failure reasons

Business questions we set out to answer

1. How many transactions are processed?
2. What is the total transaction value?
3. What are the success, failure and pending rates?
4. Which payment apps have the highest activity?
5. Which merchant categories drive the most activity?
6. Which transaction types are most common?
7. What are the major failure reasons?
8. Which apps and categories fail most often?
9. How does activity change over time?

Project Workflow



Dataset: 5,000 UPI Transactions

5,000

transaction records

CSV | UPI_Transactions Cleaned Dataset

Identity and time

Transaction_ID | Transaction_Date | Customer_Id

Location and device

City | State | Device_Type

Merchant and payment

Merchant_Category | Merchant_Name | Payment_App | Payment_Method | Transaction_Type

Amount and outcome

Transaction_Amount | Cashback_Amount | Transaction_Status | Failure_Reason

SQL analysis in MySQL

SQL concepts used

- `CREATE DATABASE`
- `CREATE TABLE`
- `SELECT`
- `WHERE`
- `GROUP BY`
- `ORDER BY`
- `COUNT`
- `SUM`
- `AVG`
- `DISTINCT`

Analysis areas

- Transaction count and value
- Success, failed, pending counts
- Average transaction amount
- Status distribution
- Payment app and category analysis
- Transaction type and failure reasons

Power Query and DAX measures

Power Query

Data preparation and transformation before modeling and dashboard development.

DAX measures

- Total Transactions
- Successful Transactions
- Failed Transactions
- Pending Transactions
- Success Rate
- Failure Rate
- Pending Rate
- Average Transaction
- Total Transaction Value
- Unique Customers

Example measures

```
Failed Transactions =
COALESCE(
  CALCULATE(
    DISTINCTCOUNT(
      UPI_Transactions[Transaction_ID]),
    UPI_Transactions[Transaction_Status]
      = "Failed"),
  0)

Failure Rate =
DIVIDE([Failed Transactions],
  [Total Transactions], 0)
```

Dashboard 1: Executive Overview

71.48%

Success rate

14.08%

Failure rate

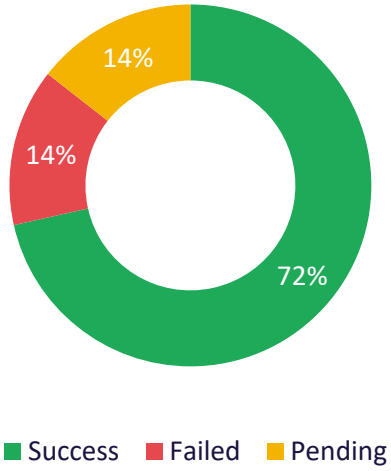
3,574

Successful transactions

704

Failed transactions

Transaction status (5,000 transactions)



Pending = total minus successful and failed. This page also covers trend, app, category and state views.

Dashboard 2: App and Merchant insights

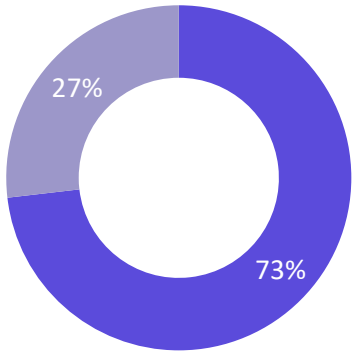
Highest app volume

Amazon Pay

1,022 transactions

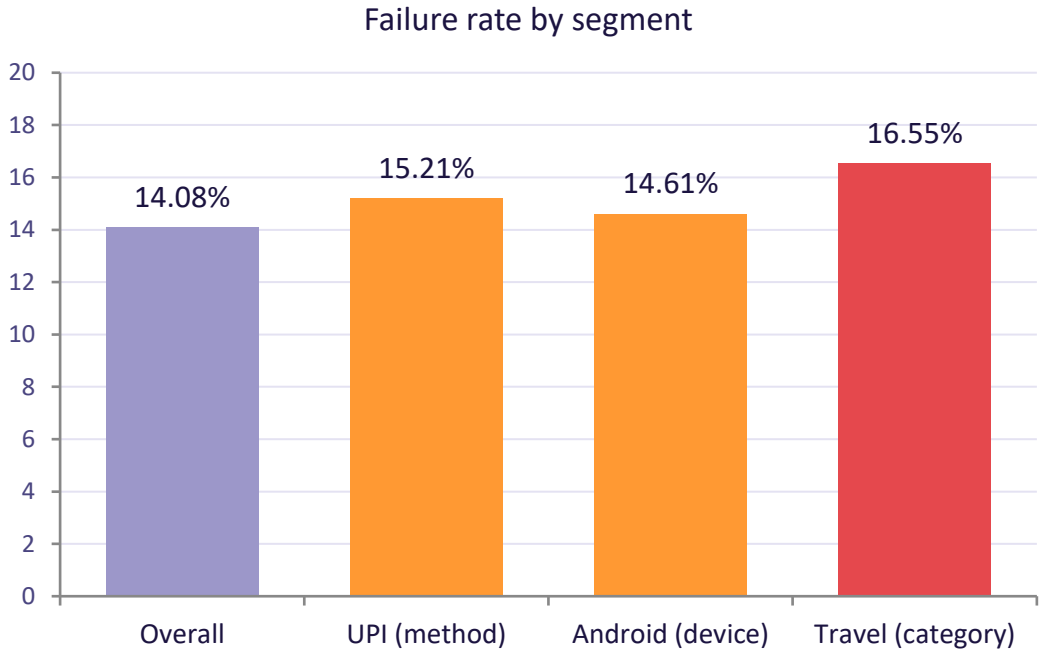
On this page: app volume and success rate, merchant category performance, transaction type and category vs average transaction.

Transaction type mix



P2M is the most common type: 3,658 transactions (73.16%).

Dashboard 3: Reliability and Operations



127

failed transactions from Bank Server Error, the top failure reason

Also on this page

Successful, failed and pending counts and rates, failure rate by app, and status by transaction type.

Key insights and Recommendations

71.48%

Overall success rate

14.08%

Overall failure rate

1,022

Amazon Pay transactions, top app

614

Transactions in March 2026, peak month

Where to focus next

- Investigate Bank Server Error, the top failure reason (127 failed transactions)
- Review Travel (16.55%), UPI (15.21%) and Android (14.61%), which sit above the 14.08% average
- Keep tracking pending transactions and monthly activity trends

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